IBM Z DATATHON 2026

Theme: Real-Time AI for Critical Decisions | Detect. Decide. Respond.

MedRelay

Catching the clinical follow-ups that fall through the cracks, as events unfold.



Evidence-aware tracking of follow-up obligations: a local LLM, a temporal evidence graph and calibrated ML, running end to end on IBM LinuxONE.

s390x big-endian | z/VM 7.3 | NNPA verified | Granite on-box

Team 4ier

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github.com/hamim-87/MedRelay

Prototype on synthetic data. Alerts are decision support for qualified human review.

THE PROBLEM

Follow-ups fall through the cracks, and “no record” is not “failed”

1 Hidden in free text

“Repeat the CBC in 48 hours. The physician must review the result. Arrange a visit in 2 weeks.” Obligations live in notes, not in structured fields.

2 Chained and fragile

A result must be reviewed before the patient is told. One missed step silently blocks every task after it.

3 Documentation lags reality

Tasks get done but are recorded late, or never. Tools that read “no record” as “failed” bury staff in false alarms.

MEASURED IN OUR SIMULATED TEST REPLAY

716 / 716

done-but-undocumented tasks flagged overdue by a naive “no record = failed” rule

0 / 716

confirmed overdue by MedRelay (flagged unverified instead)

WHO IS AFFECTED

Patients awaiting results, referrals and repeat tests

Clinicians and care coordinators drowning in alerts they cannot trust

Patient-safety teams who need an audit trail, not a guess

Why it matters: a missed result or referral can delay diagnosis, and a flood of false alarms makes real ones easier to miss. Outcomes above are simulated from explicit event timelines; no real patient data was used for follow-up outcomes.

OUR SOLUTION

A live, evidence-checked tracker for every follow-up

STEP 1 | EXTRACT

Notes become structured obligations

A local LLM plus rules propose tasks, deadlines, roles and dependencies. Every field must cite its source text, or it is rejected.

STEP 2 | TRACK

Events become verified states

A temporal evidence graph links tasks, dependencies and events. Deterministic rules decide: pending, completed, overdue, unverified, contradictory.

STEP 3 | WARN

Forecasts become explained alerts

Calibrated ML predicts which tasks will miss their deadline. Alerts say why, state their uncertainty, and go to a human.

WHAT MAKES IT DIFFERENT

The LLM never sets a state

It only proposes; rules decide.

No evidence is not failure

Unverified and censored tasks stay visible, never “failed”.

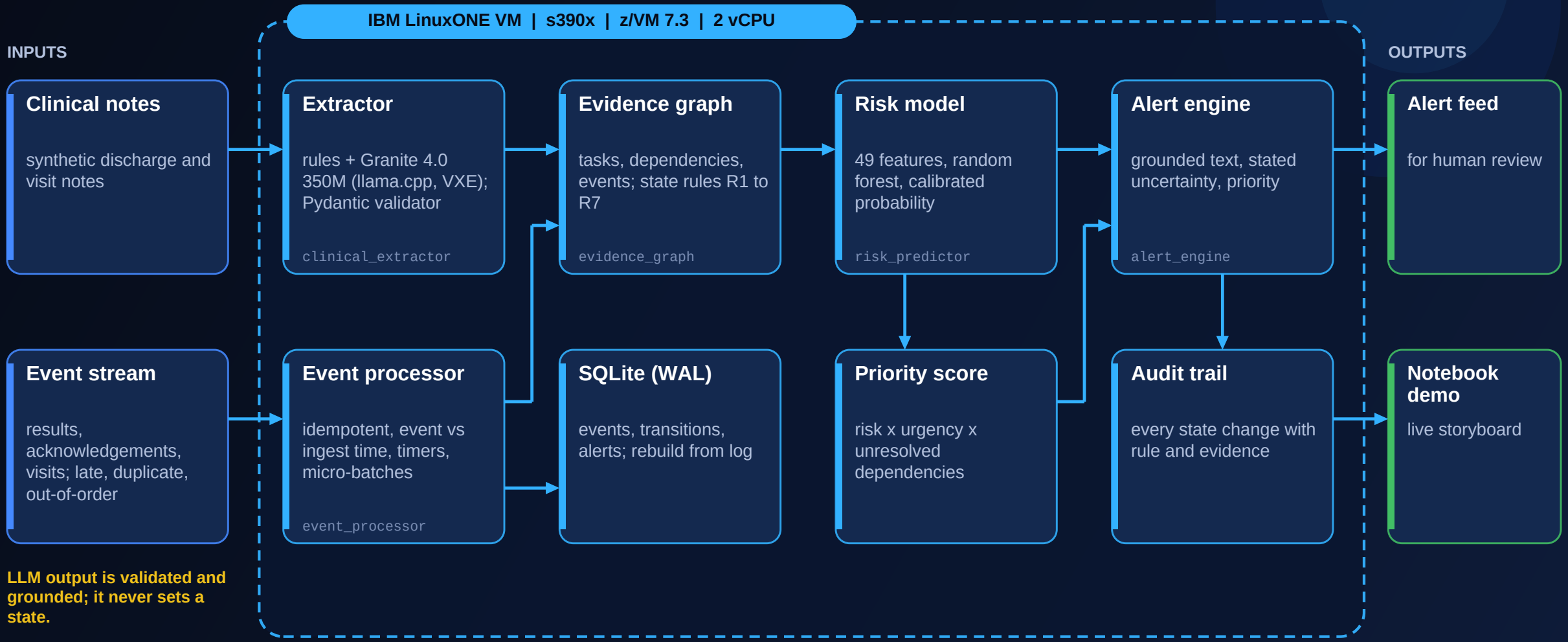
Dependency and time aware

A blocked review is flagged before the deadline.

Benefit for users: earlier warnings (median lead 72 h in our test), fewer false alarms, a full audit trail for every state change, and notes that never leave the server.

SYSTEM ARCHITECTURE

Everything runs inside one LinuxONE VM



Every box is code in the repository (src/) and was executed on the VM: clean-kernel run of the notebook, 56 code cells, 0 errors.

DATASET & MACHINE LEARNING

Leakage-safe by construction, honest about labels

3,284	24,296	34,441	10,002	110,521
patients	follow-up obligations	source events	action examples (weak labels)	real appointments (no-show)

Sources: Synthea, synthetic_hospital, MTS-Dialog, ACI-Bench (all synthetic or simulated) + Kaggle appointment no-shows (real, de-identified). Follow-up outcomes are simulated from event timelines.

DATA PIPELINE

Canonical tables
patients, encounters, notes, source events; SHA-256 checked

→

Scenario simulator
event-level timelines with documented gaps, late and missing evidence

→

Event stream
late, duplicate and out-of-order deliveries; outcomes kept in a hidden file

→

Evidence graph
103,627 nodes, 223,925 edges; dependency graph is acyclic

ML PIPELINE

49 features, 6 groups
task, own events, workload, graph dependency, graph temporal, evidence; strict event-time cutoff

→

Dense snapshots
8 assessment points per task, only while the task is still open

→

Random forest + calibration
4 model families compared on validation; split by patient; calibrated probabilities

→

Validation thresholds
alert threshold fixed on validation; applied once to test

No leakage
features see only events ingested before the assessment time

No tuning on test
thresholds and hyper-parameters chosen on validation only

Strong baselines
deadline-only rule, independent-task ML, patient-grouped bootstrap CIs

0 human-reviewed gold labels | outcomes simulated

Scores describe the simulator, not clinical accuracy. Real validation needs clinician-reviewed labels.

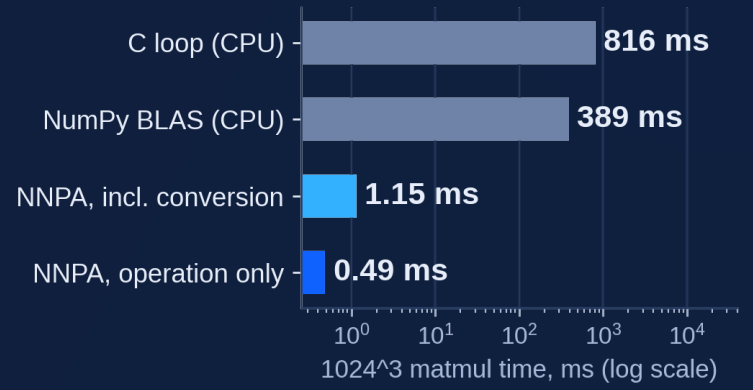
WHY IBM Z / LINUXONE

The whole stack, executed on s390x

VERIFIED ON THE VM

Platform	s390x, big-endian, z/VM 7.3.0
Resources	2 vCPU, 4.2 GB RAM (about 2.6 GB usable)
CPU facilities	vector (VXE2) and NNPA (facility 165)
Engine memory	peak 753 MB at 100,000 events (ML run)
LLM runtime	llama.cpp (VXE) + big-endian Granite 4.0 GGUF
Local inference	granite-4.0-350m-Q8_0, 15.0 tokens/s

NNPA DEVICE TEST (zDNN 1.0.1)



339x faster than NumPy BLAS on this VM incl. data conversion; max error 0.045 on values up to 13.2 (16-bit arithmetic).

20.6 k
events/s
graph + state engine (12.4 k at 100,000 events)

88 to 122
events/s
states + alerts + ML risk (random forest), 1,000 to 100,000 events

2.8 s
p99 queue latency
at 2x measured capacity, backpressure, 0 events lost

39% / 67%
events shed
bounded queue at 2x / 4x measured capacity

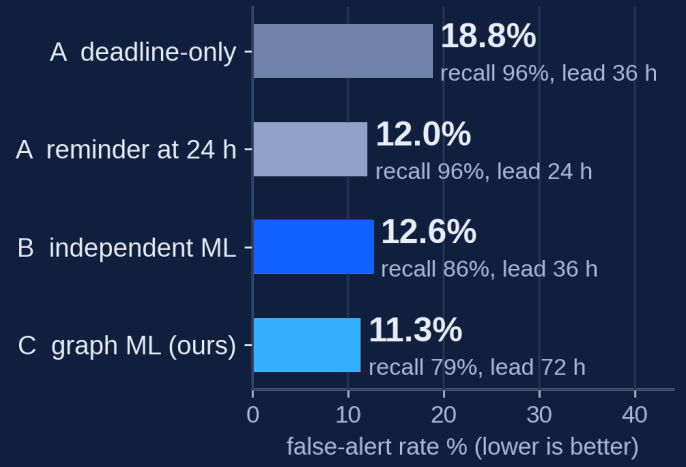
What we do not claim: the tree and linear risk models run on the CPU, not on the NNPA; only the zDNN microbenchmark proves NNPA execution. Throughput was measured on a small 2 vCPU VM; ML scoring speed depends on the model chosen on validation. No security, cost or availability claims are made.

Why LinuxONE fits: notes are processed by a local model on the same server that holds the data, and the event engine needs sustained, low-latency streaming.

EXPERIMENTS & RESULTS

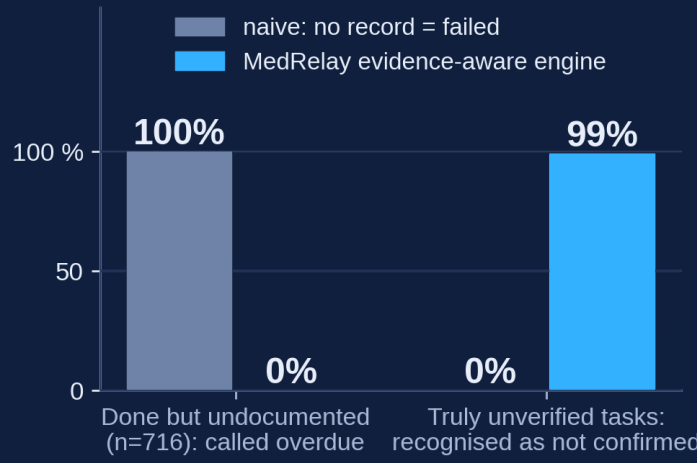
Fewer false alarms, earlier warnings, honest limits

EARLY WARNING, >= 24 h BEFORE DEADLINE



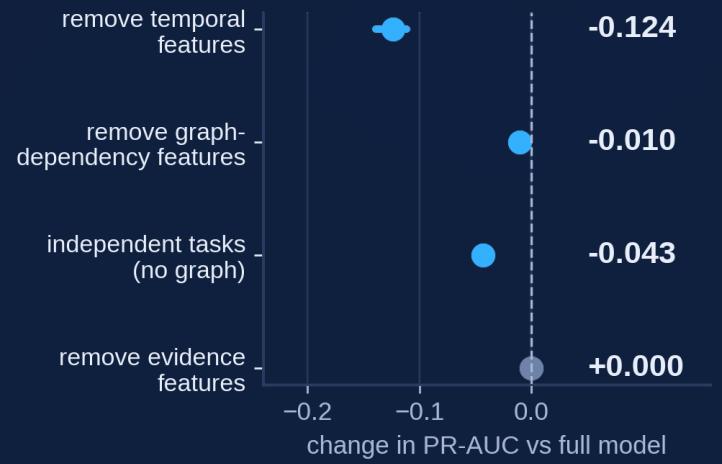
Graph model C: 1,916 alerts, 82% precision, 72 h median lead. A 24 h reminder is similar on false alerts (12.0%) with only 24 h lead; deadline-only A reaches 96% recall but alerts on 2,469 of 3,967 tasks.

EVIDENCE-AWARE STATES vs NAIVE RULE



State accuracy at the assessment time 0.911 vs 0.853; 1.000 once late evidence reconciles. The naive rule never recognises an unverified task.

WHAT DRIVES THE FORECAST



Temporal features carry most of the signal (-0.124 PR-AUC when removed). Graph features add a smaller, significant gain (+0.043 over independent tasks). Evidence features add nothing measurable.

0.893 — PR-AUC, graph model
95% CI 0.882 to 0.904; simulated outcomes

0.358 — PR-AUC, real no-show data
vs 0.203 base rate; secondary proof of concept

+0.043 — graph vs independent ML
dense-test PR-AUC, 95% CI +0.039 to +0.049; at alert level not significant (-0.001)

0.67 — failures surfaced, rules + LLM
vs 0.57 rules only; 67 failures, 120 episodes

Full-mode run on the VM (4 h 11 min): 812 test patients, 4,605 test obligations, simulated outcomes, patient-grouped bootstrap (300 resamples), thresholds chosen on validation. Over the whole window the deadline-only rule matches the ML models (max-score PR-AUC 0.963 vs 0.964); the graph model's edge is earlier: PR-AUC 0.74 vs 0.58 at 72 h ahead.

LIVE PROTOTYPE

A delayed acknowledgement, handled correctly

1 INPUT NOTE

```
Patient discharged after pneumonia
admission. Repeat the CBC within 48 hours.
The treating physician must review the
result. Contact the patient if the result
is abnormal. Arrange a follow-up
appointment in 2 weeks.
```

2 EXTRACTED OBLIGATIONS (RULES + GRANITE 350M)

ob0 **laboratory_test**
due PT48H

ob1 **followup_review**
role: ordering_clinician; after ob0

ob2 **patient_communication**
if the result is abnormal; after ob1

ob3 **appointment_referral**
due P2W; after ob1

3 AUDIT TRAIL OF THE RESULT-REVIEW TASK (VERBATIM FROM THE EXECUTED NOTEBOOK)

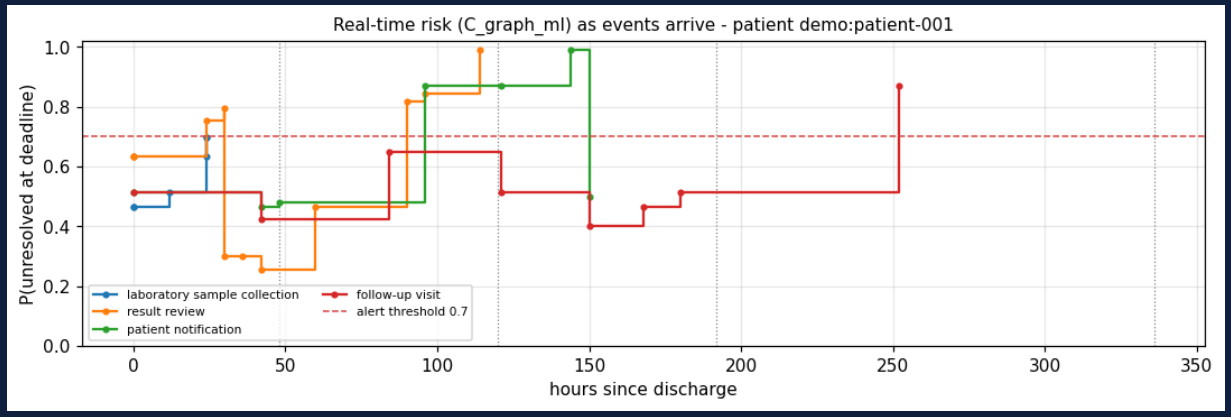
Mar 04 09:00 **pending**
R4 task created

Mar 09 09:00 **censored_not_yet_assessable**
R6 deadline passed, awaiting late evidence

Mar 09 10:00 **overdue_without_verified_completion**
R5 tracker confirms task still open

Mar 10 15:00 **completed_with_evidence**
R1 delayed acknowledgement arrives

4 RISK OF EACH TASK AS EVENTS ARRIVE (GRAPH ML, REAL OUTPUT)



Interim “overdue” was a correct state at that time and was reconciled when the late evidence arrived (verified in the notebook: overdue_without_verified_completion to completed_with_evidence). Output of notebook section 17, executed on the LinuxONE VM; the scenario events are scripted.

What is proven, and what comes next

MEASURED (FULL-MODE RUN ON LINUXONE)

0 of 716	undocumented-but-done tasks confirmed overdue (naive rule: all)
4,895	duplicate deliveries ignored; final states identical to a clean run (100%)
1.000	stored states restored identically when rebuilt from the SQLite event log
11.3% vs 18.8%	false-alert rate, graph model vs deadline-only rule, alerts >= 24 h ahead
56 / 56	notebook code cells executed without error in a clean kernel (4 h 11 min)

ANTICIPATED (NOT YET MEASURED)

Deployment	discharge follow-up monitoring fed by EHR event streams, with a human review queue
Scale-out	larger LinuxONE partitions for more patients; NNPA for neural components (current models are tree and linear)
Validation	de-identified real records and clinician-reviewed labels (there are 0 gold labels today)
Privacy	on-premises inference keeps notes on the server; a formal security evaluation is future work

vs reminder systems

evidence-aware: absence is unverified, not failed

vs LLM-only pipelines

validated, grounded fields; deterministic states

vs risk scores alone

dependency and time aware, with an audit trail

CONCLUSION

No follow-up left behind

0 / 716

false failures

Evidence-aware states never call undocumented work “failed”; the naive rule did, every time.

11.3%

false-alert rate

Graph forecasting warns 72 h ahead with fewer false alerts than a deadline-only rule (18.8%).

s390x

end to end

Local Granite LLM, streaming engine and ML, plus a verified NNPA device test, all on one LinuxONE VM.

WHY LINUXONE MATTERS HERE

Sensitive notes are processed by a local model on the server that holds the data.

The same VM runs extraction, graph, ML and the stream engine, with recoverable state.

NNPA execution is verified, a path for future neural models.

NEXT STEPS

Clinician-reviewed labels to replace weak and simulated ones.

Validate on de-identified real records under governance.

Add neural components that can use the NNPA, and test at larger scale.

Detect. Decide. Respond. And never mistake silence for failure.

Code and notebook: github.com/hamim-87/MedRelay

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